Race prediction works. Now make useful representations deepen.

Our ambition spans language and reasoning, multimodal world models, embodiment, continual learning and specialized hardware. This demonstration connects one measured language result to the next scaling proof.

Measured: better language prediction on the same 4M-token development recipe.

0.3749 nats less loss per token. Lower is better. Both models score the same 8,192 development targets in 1,024-token windows; one seed and different model sizes. Bars start at zero.

Why this matters

The winning model predicts through competing causes: earlier text, continuing spans, learned concept units and a background route. Its registered main-loss operator subtotal is 7.74 million FLOPs per target; auxiliary-loss backward, optimizer and unregistered work are excluded. This is evidence that our prediction law can compete in language.

Where the advantage comes from

Tokens already seen in the window: **3.16 versus 3.75** NLL (race versus dense).

Tokens new to the window: **7.87 versus 8.02** NLL (race versus dense).

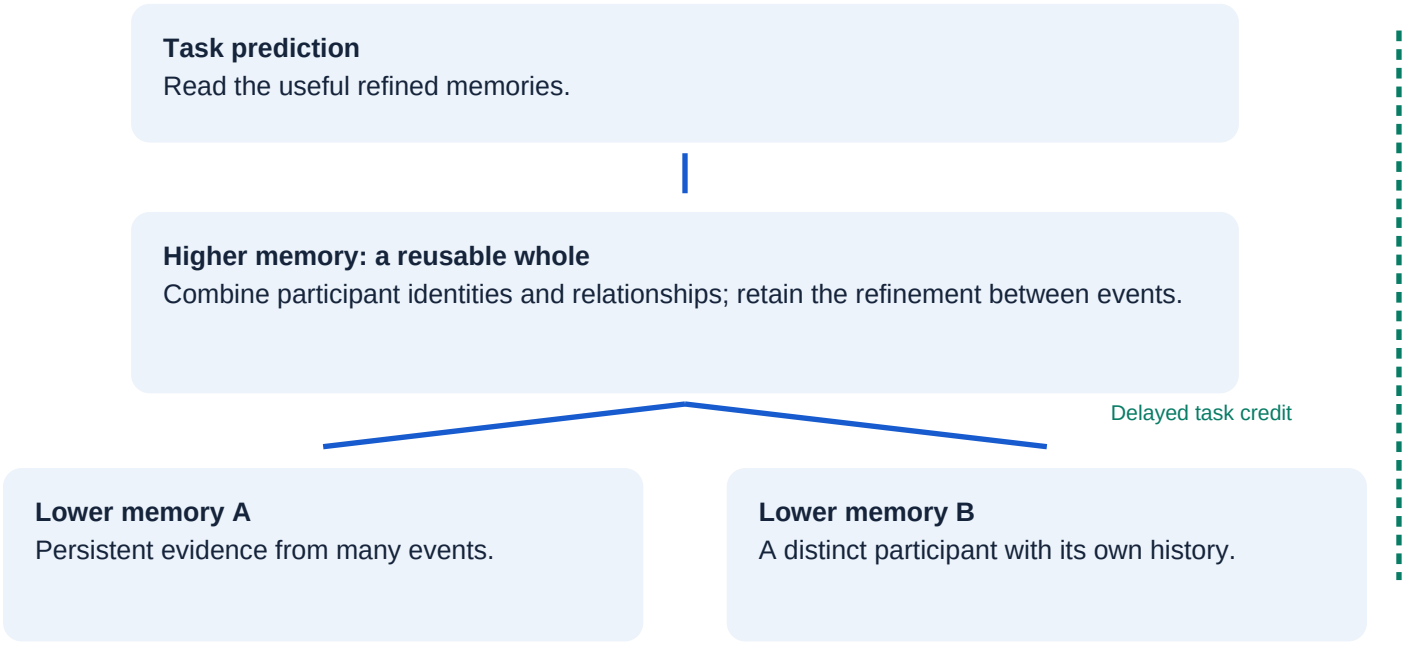
These are descriptive subsets of the saved predictions. Remembered evidence supplies a strong foothold; novel prediction and relations are the opportunity for deeper composition.

Scope of the win

The winner uses two temporal-memory carrier layers and BPTT. The fully native eight-layer event-local learner is a separate development track. This is a development comparison, separate from the 13 public benchmark wins.

Memories hold meaning. Events schedule refinement.

Higher stages query several lower memories, compose their relationships, and write persistent refinements. Later task success must teach the producers of those memories.



Architecture schematic; learned semantic hierarchy is the next proof. Events can arrive at different rates; a persistent memory can remain useful between updates.

Implemented now
Bounded credit replay follows the actual lower-memory versions used by a historical composer. It adds no parameters and is designed to preserve factual predictions. Its additional replay is capped at six child reads per root and two edges per path.

Numerical validation: passed.
The gate checks an independent finite-computation gradient, unchanged predictions and RNGs, bounded work, and update/recovery on a private copy of the acquired eight-layer learner.

The investor proof milestone
Deeper refinements must improve held-out prediction and relations at disclosed total learning work. Larger memory must produce useful quality while activity, discovery, traffic and storage are measured. Independent firing must save actual work while retaining quality.

Existing depth/capacity experiments are running; composition-credit validation is queued. Historical route/timing and some input provenance remain outside the new bounded credit path. No new scaling or hardware-energy result is claimed.

Sources: composition_demonstration_evidence.json binds the original results, error analysis, theory, implementation and immutable gate admission. Use the accompanying evidence brief for the full public benchmark record.